PROJECT OVERVIEW · PLAIN LANGUAGE

Harbour Twin

A 3D computer copy of a harbour, built only from free public data, that tracks ships, finds the ones that are not announcing themselves, plans for floods, and checks its own accuracy.

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01 In one minute

A **digital twin** is a working computer copy of a real place. Harbour Twin is a 3D copy of Vancouver Harbour (and St. John's Harbour) made **only from free public data**: laser scans of the ground, satellite photos and radar, building outlines, tide gauges and the radio messages ships broadcast about where they are (**AIS**).

It answers three practical questions, and shows its working for each:

- **Which ships are out there, and are any of them silent?** It compares ships seen by satellite radar with ships announcing themselves over AIS. A ship the radar sees but that is not announcing itself is worth a closer look.
- **Where would the sea reach in a flood?** It works out, building by building, the water level at which the sea actually reaches it, not just "everything below a line".
- **Can the data be trusted?** Every dataset is scored against the most accurate one available (laser scans) and must pass published accuracy specifications before it is used.

77
ships found by satellite radar in one Vancouver pass, including the anchored freighters in English Bay

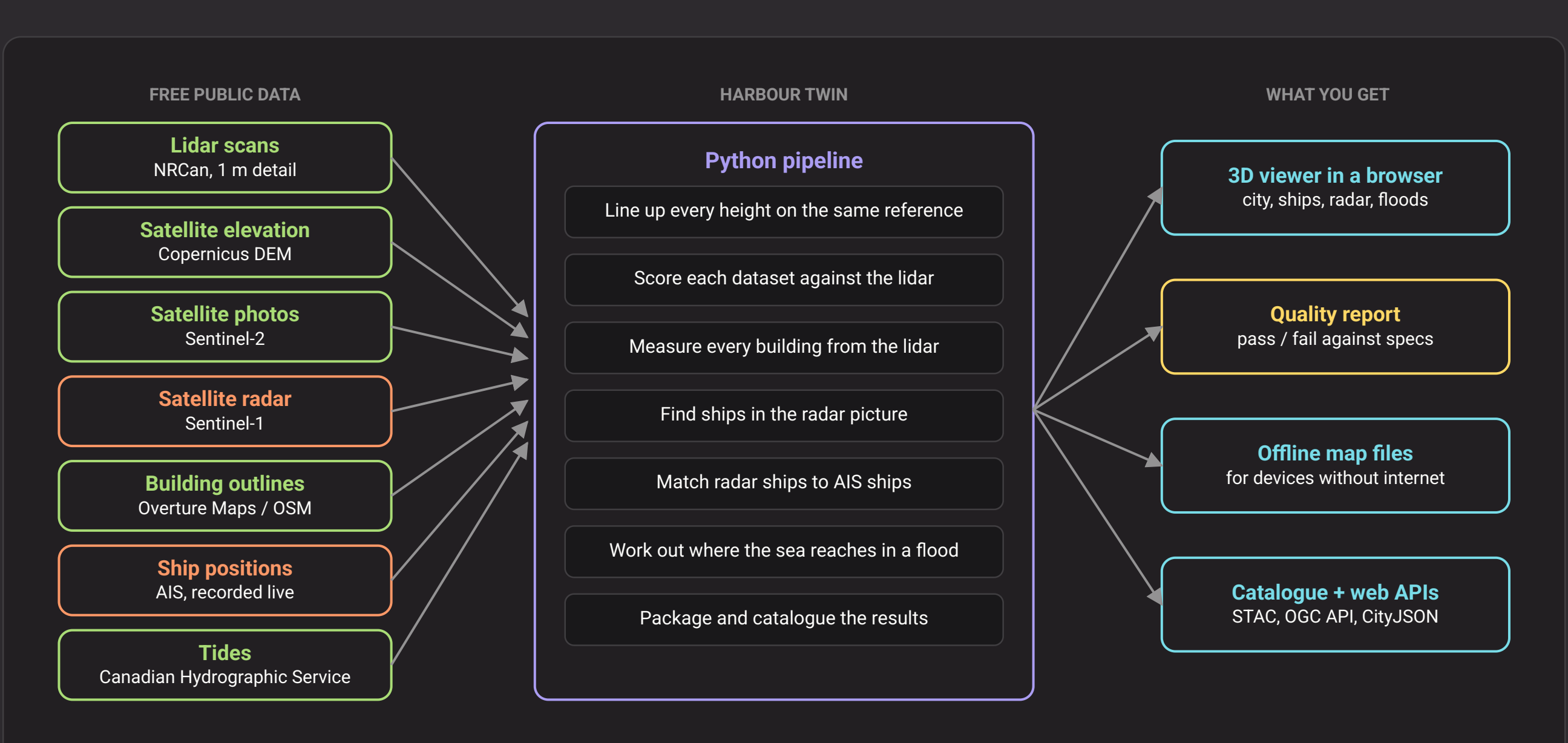
347
vessels recorded live over 4 hours: 7,854 cleaned position reports

98.7%
of 5,848 building heights measured by laser, not guessed

67
automated checks that must pass before results are published

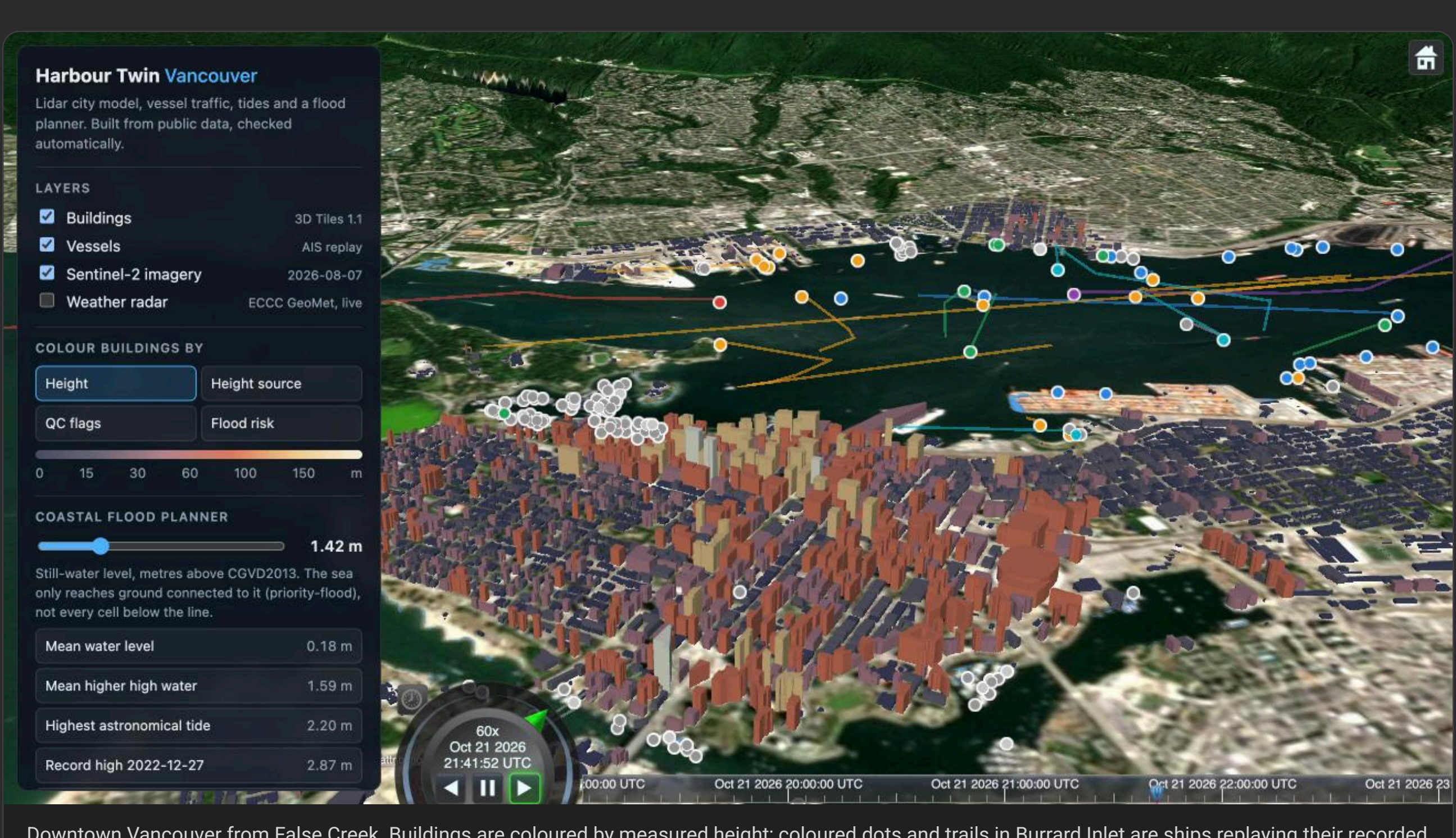
02 The big picture

Public data goes in on the left, a checked set of results comes out on the right.


FREE PUBLIC DATA
Lidar scans
NRCan, 1 m detail
Satellite elevation
Copernicus DEM
Satellite photos
Sentinel-2
Satellite radar
Sentinel-1
Building outlines
Overture Maps / OSM
Ship positions
AIS, recorded live
Tides
Canadian Hydrographic Service
HARBOUR TWIN
Python pipeline
Line up every height on the same reference
Score each dataset against the lidar
Measure every building from the lidar
Find ships in the radar picture
Match radar ships to AIS ships
Work out where the sea reaches in a flood
Package and catalogue the results
WHAT YOU GET
3D viewer in a browser
city, ships, radar, floods
Quality report
pass / fail against specs
Offline map files
for devices without internet
Catalogue + web APIs
STAC, OGC API, CityJSON

Orange inputs are the maritime ones: satellite radar and ship positions. Everything is read directly from the organisations that publish it; nothing is copied by hand.

03 What it looks like



Harbour Twin Vancouver

Lidar city model, vessel traffic, tides and a flood planner. Built from public data, checked automatically.

LAYERS
- ☑ Buildings — 3D Tiles 1.1
- ☑ Vessels — AIS replay
- ☑ Sentinel-2 imagery — 2026-08-07
- ☐ Weather radar — ECCC GeoMet, live

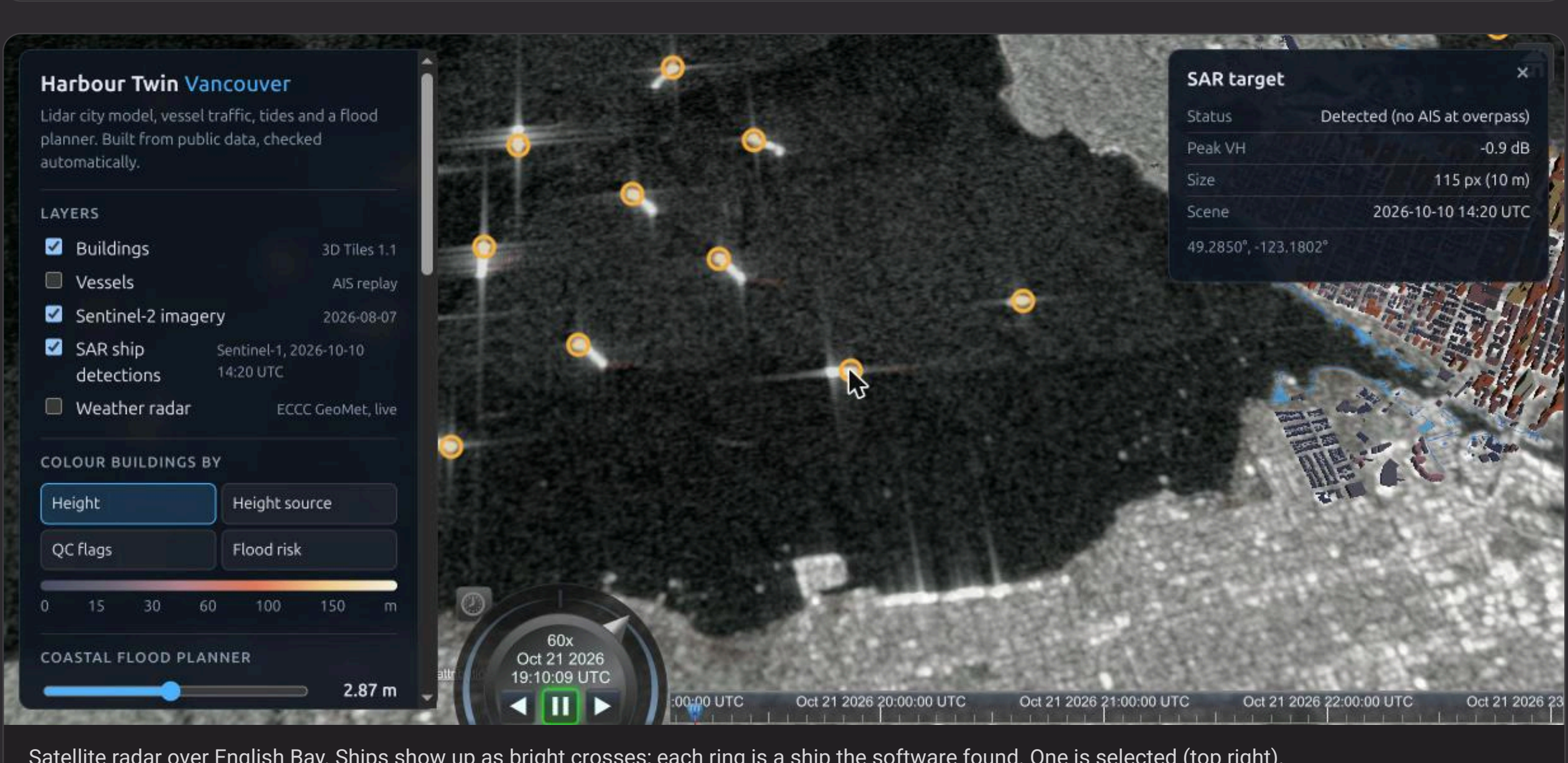
COLOUR BUILDINGS BY: Height (selected)

COASTAL FLOOD PLANNER: 1.42 m

Still-water level, metres above CGVD2013. The sea only reaches ground connected to it (priority-flood), not every cell below the line.

Mean water level	0.18 m
Mean higher high water	1.59 m
Highest astronomical tide	2.20 m
Record high 2022-12-27	2.87 m

Downtown Vancouver from False Creek. Buildings are coloured by measured height; coloured dots and trails in Burrard Inlet are ships replaying their recorded movements, with the water level following the real tide on the same clock.



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- ☑ Sentinel-2 imagery — 2026-08-07
- ☑ SAR ship detections — Sentinel-1, 2026-10-10 14:20 UTC
- ☐ Weather radar — ECCC GeoMet, live

COLOUR BUILDINGS BY: Height (selected)

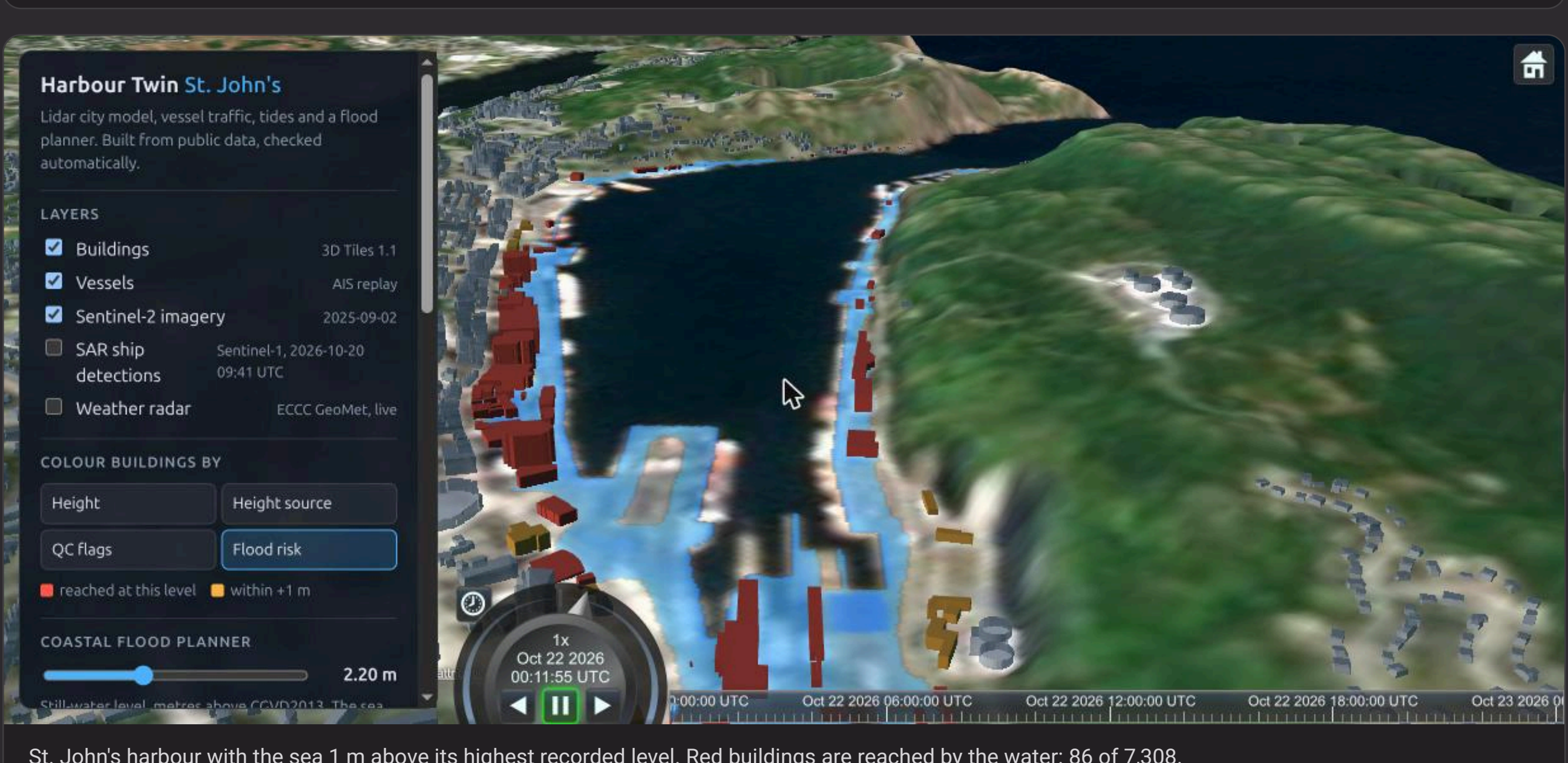
COASTAL FLOOD PLANNER: 2.87 m

SAR target

Status	Detected (no AIS at overpass)
Peak VH	-0.9 dB
Size	115 px (10 m)
Scene	2026-10-10 14:20 UTC

49.2850°, -123.1802°

Satellite radar over English Bay. Ships show up as bright crosses; each ring is a ship the software found. One is selected (top right).



Harbour Twin St. John's

Lidar city model, vessel traffic, tides and a flood planner. Built from public data, checked automatically.

LAYERS
- ☑ Buildings — 3D Tiles 1.1
- ☑ Vessels — AIS replay
- ☑ Sentinel-2 imagery — 2025-09-02
- ☐ SAR ship detections — Sentinel-1, 2026-10-20 09:41 UTC
- ☐ Weather radar — ECCC GeoMet, live

COLOUR BUILDINGS BY: Flood risk (selected)

🟥 reached at this level 🟨 within +1 m

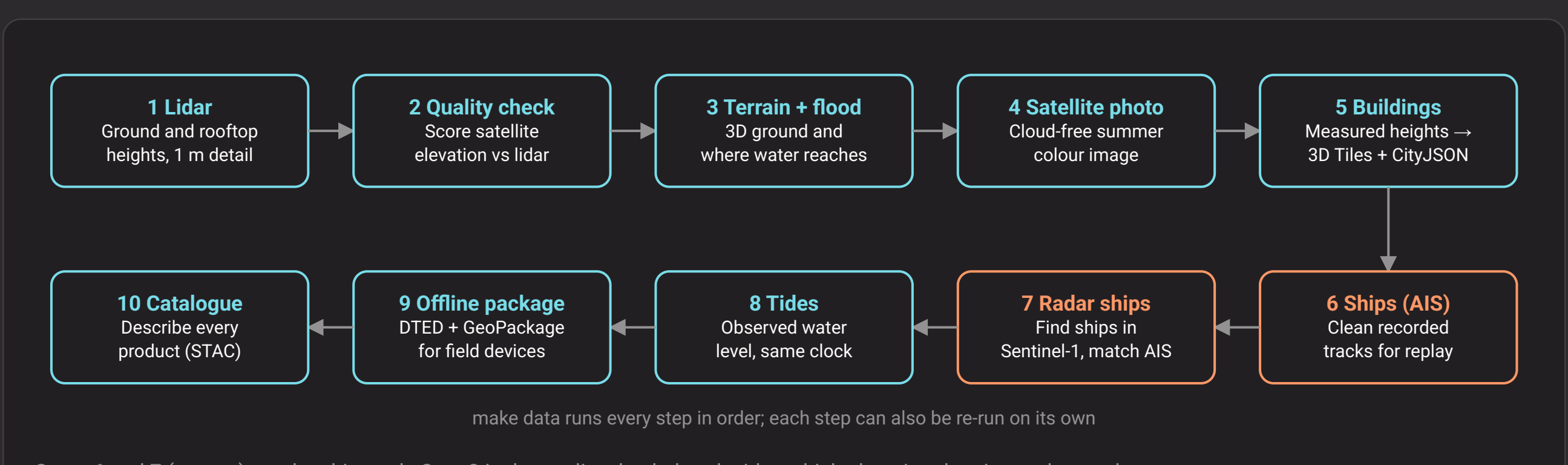
COASTAL FLOOD PLANNER: 2.20 m

St. John's harbour with the sea 1 m above its highest recorded level. Red buildings are reached by the water: 86 of 7,308.

A 30-second tour is in the folder as `demo.mp4`.

04 How a run works

One command builds the whole twin, step by step. Each step saves its results, so any step can be re-run alone.

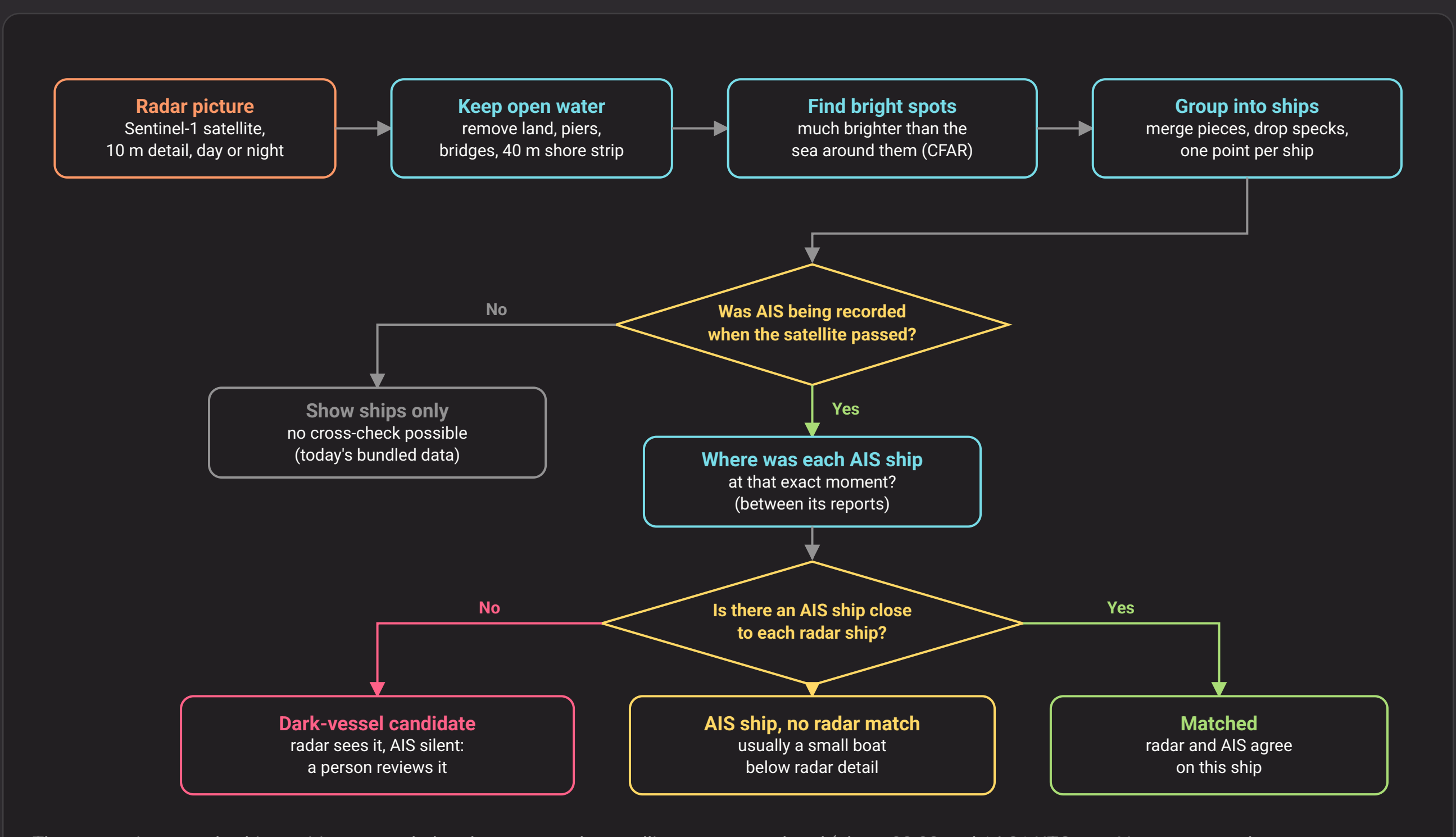

1 Lidar
Ground and rooftop heights, 1 m detail
2 Quality check
Score satellite elevation vs lidar
3 Terrain + flood
3D ground and where water reaches
4 Satellite photo
Cloud-free summer colour image
5 Buildings
Measured heights → 3D Tiles + CityJSON
6 Ships (AIS)
Clean recorded tracks for replay
7 Radar ships
Find ships in Sentinel-1, match AIS
8 Tides
Observed water level, same clock
9 Offline package
DTED + GeoPackage for field devices
10 Catalogue
Describe every product (STAC)

make data runs every step in order; each step can also be re-run on its own

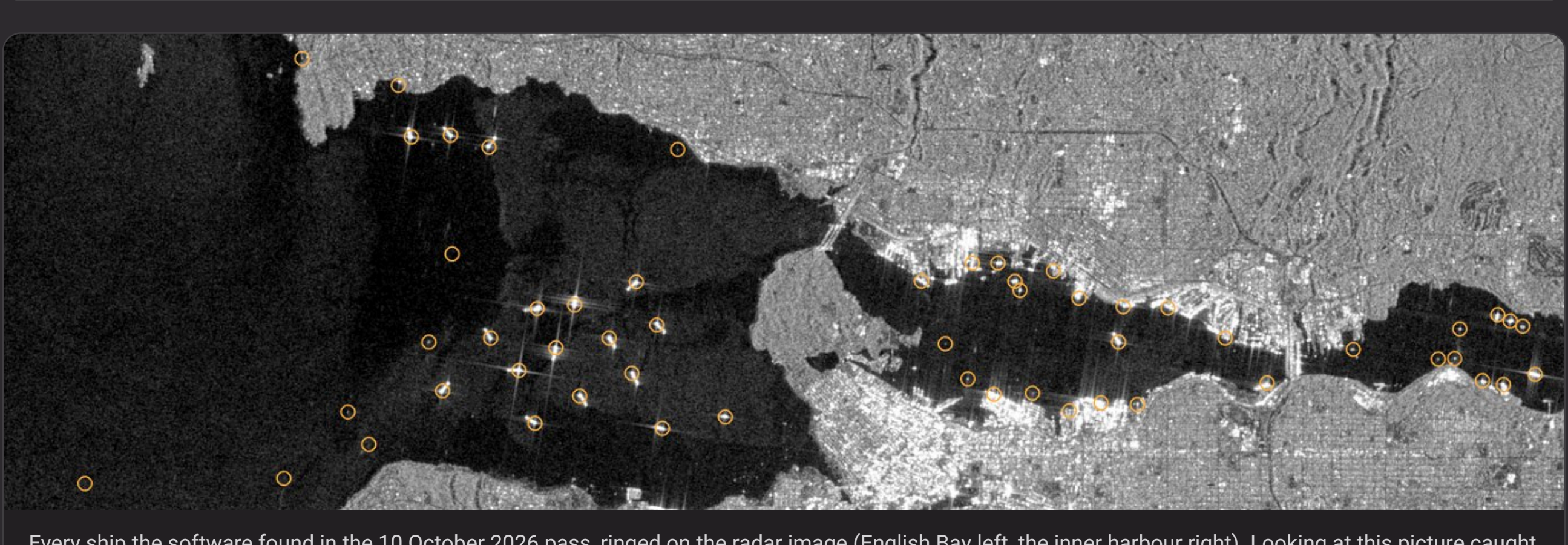
Steps 6 and 7 (orange) are the ship work. Step 2 is the quality check that decides which elevation data is good enough to use.

05 Finding ships that do not announce themselves

Large ships must broadcast AIS: who they are and where they are, every few seconds. Some don't: their equipment is off, broken, or deliberately switched off. Satellite radar sees ships whether or not they broadcast, day or night, through cloud. Comparing the two is a core task in **maritime domain awareness**.


Radar picture
Sentinel-1 satellite, 10 m detail, day or night
Keep open water
remove land, piers, bridges, 40 m shore strip
Find bright spots
much brighter than the sea around them (CFAR)
Group into ships
merge pieces, drop specks, one point per ship
Was AIS being recorded when the satellite passed?
No
Show ships only
no cross-check possible (today's bundled data)
Yes
Where was each AIS ship at that exact moment? (between its reports)
Is there an AIS ship close to each radar ship?
No
Dark-vessel candidate
radar sees it, AIS silent: a person reviews it
AIS ship, no radar match
usually a small boat below radar detail
Yes
Matched
radar and AIS agree on this ship

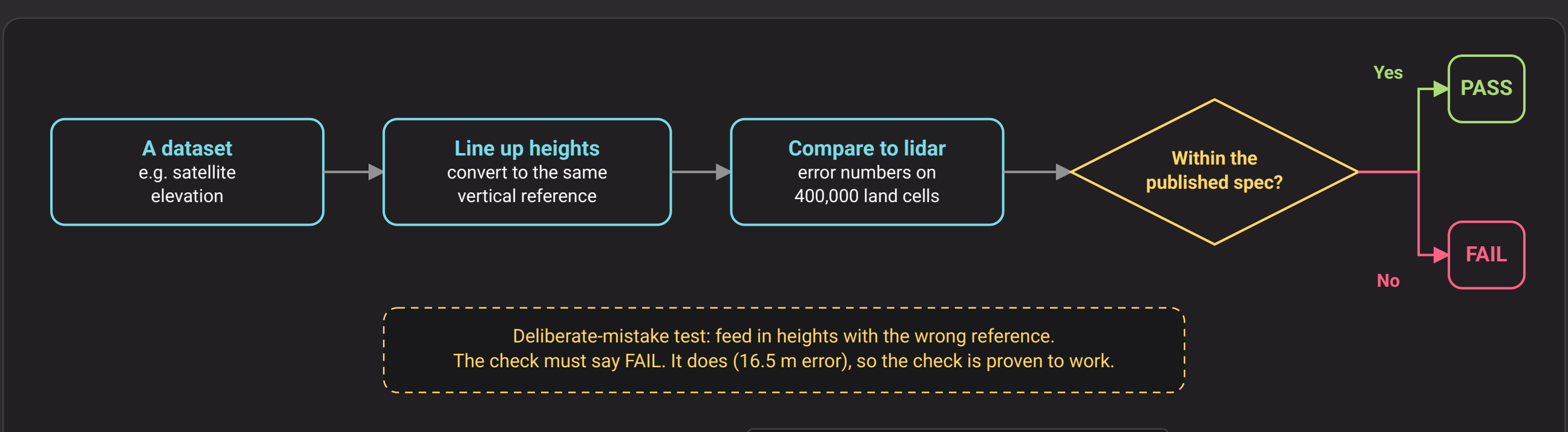
The comparison needs ship positions recorded at the moment the satellite passes overhead (about 02:02 and 14:21 UTC over Vancouver, each repeating every 12 days). The software predicts the next pass.



Every ship the software found in the 10 October 2026 pass, ringed on the radar image (English Bay left, the inner harbour right). Looking at this picture caught a mistake: the Lions Gate Bridge had been counted as a ship. Bridges are now excluded.

Honest status. The ship finding works on real radar images. The matching step is tested with made-up data so far, because the bundled ship recording does not overlap a satellite pass. Recording ships during the next pass (path D below) produces the first real comparison.

06 How we know the results are right

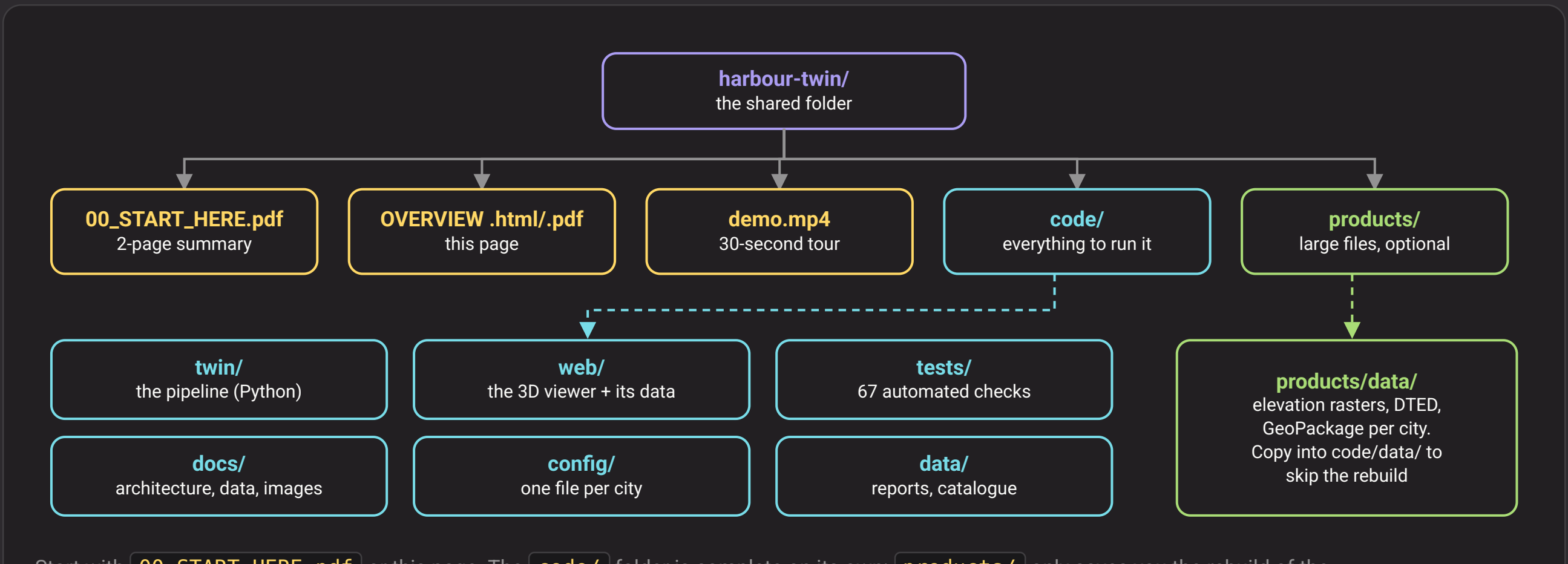

A dataset
e.g. satellite elevation
Line up heights
convert to the same vertical reference
Compare to lidar
error numbers on 400,000 land cells
Within the published spec?
Yes
PASS
No
FAIL
Deliberate-mistake test: feed in heights with the wrong reference.
The check must say FAIL. It does (16.5 m error), so the check is proven to work.

Every elevation dataset goes through this before it is used. The results are in `code/data/vancouver/QC_REPORT.md`.

What the checks caught

- **A standard tool was wrong by up to a metre.** The common mapping library (PROJ) converts Canadian heights to the global reference with an error of 0.30 m in Vancouver and 1.05 m in St. John's. The pipeline does the conversion explicitly, and a test makes sure it stays right.
- **Published building heights are often wrong.** A 190.8 m tower was listed as 10 m tall in OpenStreetMap. 717 such disagreements are flagged for a person to review rather than silently "fixed".
- **A bridge looked like a ship** to the radar detector, found by looking at the results on the radar image.
- **A second city broke two assumptions**: a missing software dependency, and a tide gauge in St. John's that does not publish the usual height reference. Its level is now converted another way, and cross-checked.

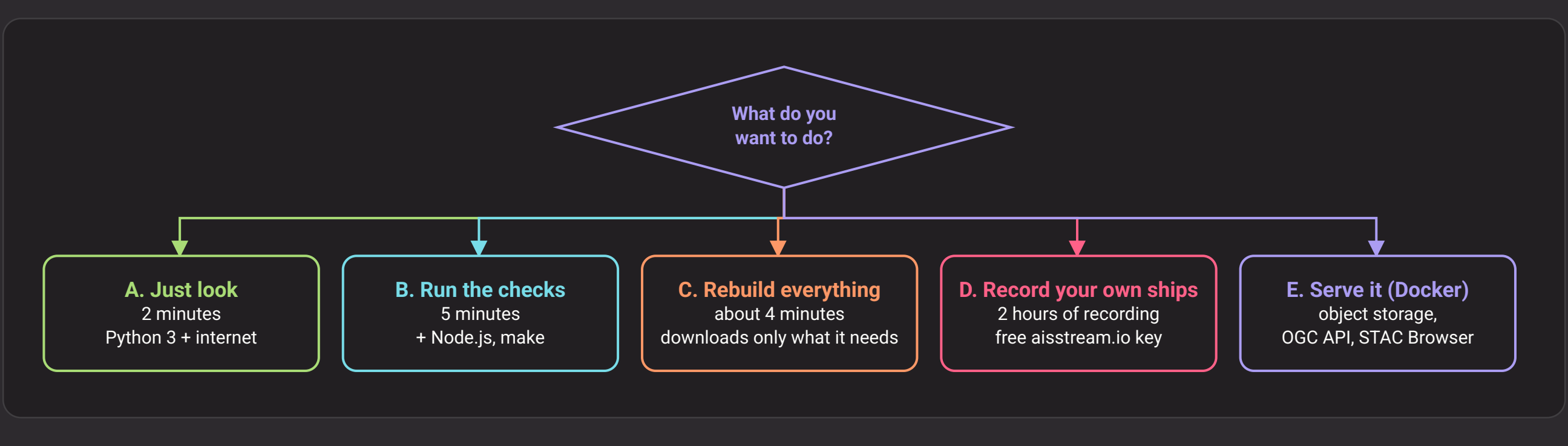
07 What is in the folder



Start with `00_START_HERE.pdf` or this page. The `code/` folder is complete on its own; `products/` only saves you the rebuild of the large files.

08 Reproduce it yourself

Pick the path that matches what you want. Each one builds on the ones before it.



Before you start: download the whole `harbour-twin` folder (Google Drive gives you a .zip; unzip it). Open a terminal in the unzipped folder. Linux and macOS work directly; on Windows, use WSL (Windows Subsystem for Linux).

A. Just look at it

Needs: Python 3 (any recent version) and an internet connection for the 3D library and base map.

```
cd code
python3 -m http.server 8765 --directory web
```

Open http://127.0.0.1:8765 in Chrome, Edge or Firefox. For St. John's open `http://127.0.0.1:8765/?aoi=stjohns`. Stop it with `Ctrl+C`.

Expect: downtown Vancouver in 3D, ships moving in the harbour, a panel on the left with layers, flood planner and data-quality numbers. Tick "SAR ship detections" to see the radar layer.

Opening `web/index.html` by double-clicking does not work: browsers block a page from reading its own data files unless it is served, which is what the first command does.

B. Run the checks

Needs: Python 3.11 or newer, Node.js 18 or newer, and `make`.

```
cd code
make setup        # installs Python packages and the 3D-model validator (about 30 s)
make test         # the automated checks
make validate     # checks every 3D building tile against the official standard
```

Expect: `64 passed, 3 skipped`, then `{"files":69,"errors":0,"warnings":0}`. The 3 skipped checks need the large files. To include them, copy them in and run the checks again:

```
cp -r ../products/data/* data/
make test         # now 67 passed
make serve        # same viewer as path A, using the installed Python
```

C. Rebuild everything from the original sources

Needs: path B done, and a normal internet connection. It reads only the parts of each dataset it needs, not whole files.

```
make data                 # Vancouver: every step, about 4 minutes
make test validate        # the same checks, now on what you just built
make data AOI=stjohns     # St. John's, same pipeline, different settings file
make test AOI=stjohns
```

Expect: progress messages for each step ("step lidar done in 13 s" and so on), then the same checks passing. Numbers can move slightly if a source publishes newer data.

D. Record your own ships, and compare them with radar

Needs: path B done and a free key from aisstream.io (sign in, copy the API key).

```
echo "AISSTREAM_API_KEY=paste-your-key-here" > .env
make passes               # when the radar satellite next passes over
make record MIN=120       # start about an hour before that pass; records 2 hours
make ais                  # cleans the tracks, finds radar ships, matches them
```

Expect: after the satellite image appears in the public catalogue (usually 1 to 3 days after the pass), `make ais` reports how many radar ships were matched, how many are dark-vessel candidates, and how many AIS ships the radar did not see. The viewer's SAR panel shows the same.

Recording adds to any earlier recording. To keep them apart, move the old one first: `mv data/cache/vancouver/ais_raw.jsonl data/cache/vancouver/ais_raw_old.jsonl`.

E. Serve it like a product (Docker)

Needs: the large files in `data/` (path B or C) and Docker with the Compose plugin.

```
docker compose up --build        # first run downloads the images (a few minutes)
```

Expect, once the publish step has exited: the viewer at `http://localhost:8333/harbour-twin/web/index.html`, served from object storage; the data as a web service (OGC API) at `http://localhost:5000`; and the catalogue at `http://localhost:8080`. Stop with `Ctrl+C`, then `docker compose down`.

If something goes wrong

What you see	What it means and what to do
Could not resolve host	A short network hiccup reaching one of the data providers. Run the same command again; every step can be re-run safely.
3 skipped	The large files are not in `data/`. Copy them from `products/` (path B) or rebuild them (path C).
Blank viewer page	The page was opened by double-clicking. Use the `python3 -m http.server` command from path A.
make: command not found	Install it (`sudo apt install make` on Ubuntu; Xcode command-line tools on macOS), or use WSL on Windows.
npm: command not found	Install Node.js 18+ from nodejs.org. It is only needed for `make validate`.
Set AISSTREAM_API_KEY	Only recording needs a key (path D). Everything else works without one.

09 Limits worth knowing

- The flood planner is a planning screen: it uses still water and the shape of the ground, not waves, wind or drainage.
- The radar matching has not yet been run on a real ship recording made during a satellite pass (see section 5).
- How bright a spot must be to count as a ship was set by inspecting real images, not from a labelled training set.
- AIS comes from volunteer receivers, so coverage is uneven and many vessels never report their type.
- Buildings newer than the laser scan fall back to published heights, and are labelled as such.

10 Words used here

AIS

Automatic Identification System: radio messages in which ships report their identity, position, speed and course.

SAR (radar)

Synthetic Aperture Radar: a satellite sends microwaves and records the echo. Works at night and through cloud; metal ships echo brightly against a dark sea.

Sentinel-1 / -2

European Space Agency satellites: Sentinel-1 carries radar, Sentinel-2 takes colour images. Their data is free.

Lidar

Laser scanning from aircraft. Measures the ground and rooftops to within centimetres; the most accurate height data used here.

Dark vessel

A ship seen by sensors such as radar but not reporting over AIS. Here always a *candidate*, for a person to review.

CFAR

Constant False Alarm Rate: a standard radar method that calls a spot a target only if it is much brighter than its own surroundings.

Vertical datum

The "zero" that heights are measured from. Different datasets use different zeros, metres apart; mixing them is a common, silent error.

Digital twin

A working computer copy of a real place, built from measurements and kept checkable.

STAC

SpatioTemporal Asset Catalog: a standard way to describe and find map and satellite data; used here both to find inputs and describe outputs.

DTED, GeoPackage

Standard map-file formats used on devices that work without internet, such as ships and drones.

CityJSON

An open format for 3D city models (the JSON form of CityGML), read by city-planning tools. The buildings are exported to it as simple closed blocks.

OGC API - Features

A standard web interface for map data: any GIS can ask it for "buildings in this box" or "flagged buildings" and get them back.

Object storage

Cloud file storage such as Amazon S3. Publishing the products there lets anyone read them over the web without a special server.

3D Tiles

An open standard for streaming 3D city models to a web browser.

Data: NRCan (Open Government Licence – Canada) · Copernicus DEM © DLR/Airbus · Copernicus Sentinel-1 and Sentinel-2 (ESA; Sentinel-1 RTC via Microsoft Planetary Computer, CC BY 4.0) · Overture Maps (ODbL / CDLA) · © OpenStreetMap contributors · aisstream.io · Fisheries and Oceans Canada (CHS) · ECCC · Open-Meteo · CARTO. Code: PolyForm Noncommercial 1.0.0 (noncommercial use only; data keeps its sources' licences). Built with an AI coding agent (Claude Code) under the author's direction; every result checked by automated tests and by review. Technical detail: `code/docs/ARCHITECTURE.md`.